

Lab Assignment 1

RTL Design and Functional Synthesis using Verilog

Course: Synthesis and Optimization of VLSI Circuits

Assignment Title: RTL Coding for Functional Synthesis

Objective:

The objective of this assignment is to strengthen understanding of **Register Transfer Level (RTL) design** and **functional synthesis** using Verilog HDL. Students are required to design synthesizable Verilog modules that correctly implement the given specifications.

All designs must follow **synthesis-friendly RTL coding guidelines**.

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Instructions

1. Write synthesizable Verilog HDL code for each problem.
 2. Do **not** use delays (`#`), `initial` blocks, or non-synthesizable constructs.
 3. Clearly specify reset type (synchronous/asynchronous) as mentioned.
 4. Simulate and verify functionality using a suitable testbench.
 5. Include waveform screenshots and brief explanation for each module.
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Problem 1: 4-bit Synchronous Up Counter

Design a **4-bit synchronous up counter** with the following specifications:

- Active-high synchronous reset
- Enable input
- Counter increments on every rising edge of the clock when enable is high

- Counter resets to zero when reset is asserted

Problem 2: 8-bit Register with Load and Clear

Design an **8-bit register** with:

- Load control signal
- Active-low asynchronous clear
- On load assertion, the register should load input data
- Otherwise, the register should retain its previous value

Problem 3: 4:1 Multiplexer

Design a **4-to-1 multiplexer** using a `case` statement:

- Four single-bit inputs
- Two-bit select line
- One single-bit output

Problem 4: 4-bit Magnitude Comparator

Design a **4-bit magnitude comparator** that compares two 4-bit numbers A and B and generates the following outputs:

- $A > B$
- $A = B$
- $A < B$

Problem 5: Mealy FSM – Sequence Detector (1011)

Design a **Mealy finite state machine** that detects the overlapping sequence **1011** from a serial input stream.

- Output should be asserted when the sequence is detected
- FSM should allow overlapping sequences

Tools Required

- Xilinx Vivado
- Verilog HDL

Task Description

1. Write Verilog code.
2. Verify functionality using simulation.
3. Perform RTL synthesis in Vivado.
4. Analyze RTL schematic and synthesized netlist.

Expected Outcome

- Functionally correct RTL design.
- Successful synthesis with generated RTL and technology schematic.

Evaluation Scheme (15 Marks)

- Verilog coding correctness – 5
- Functional simulation results – 4
- RTL & synthesized schematic analysis – 3
- Viva / explanation – 2
- Quality of simulation and documentation-1